

What Makes a Good Wine?

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Background

The wine industry is a blend of agriculture, science, art and business. The industry is made up of thousands of vineyards around the globe, with each wine-producing country or region having its own unique climates and soil types. These factors, along with the type of grape grown and how the wine was aged, can contribute significantly to the taste and quality of the beverage. The sheer variety and complexity of available vintages makes it challenging to confidently select a high-quality wine.

To aid consumers in that decision, this paper uses machine-learning methods to create a model to predict a wine's quality based on its chemical make-up. It also examines the characteristics of a high-quality wine.

Data Set

To model the wine's quality as determined by its chemical characteristics, this paper utilized data sets on wine quality donated to the UC Irvine Machine Learning Repository.¹ The donated data include two data sets of red and white wines sampled from northern Portugal from 2024 to 2007 by the local wine certification agency, CVRVV. Across the two data sets there were 6,497 vintages sampled. For privacy reasons, the data do not include grape type, wine brand, selling price, or vineyard name.

Both data sets included 11 quantitative variables to measure certain chemical aspects for each wine sampled — such as the amount of sulfates, alcohol level, or pH of each wine. The data set also included 1 categorical, ordinal variable measuring the quality of the wine. To determine the quality, at least 3 taste testers evaluated each wine on a scale of 1 to 10. The final quality score is the median of all the taste testers' quality scores. As a result, the lowest level of quality is scored as 3 and the highest is scored as a 9. Quality is the dependent variable in the following prediction models.

Exploratory Data Analysis

Below is a table with the summary statistics and the results of a Shapiro-Wilks test for normality. Based on the Shapiro-Wilks test, all of the variables are normally distributed — including the categorical variable “quality.” Therefore, “quality” was treated as numeric value in the following cluster analysis.

Variable	Mean	Median	SD	Min	Max	Shapiro-Wilks P-Value
quality	3.818	4.000	0.873	1.000	7.000	0.000
fixed.acidity	7.215	7.000	1.296	3.800	15.900	0.000
volatile.acidity	0.340	0.290	0.165	0.080	1.580	0.000

¹Cortez, P., Cerdeira, A., Almeida, F., Matos, T., & Reis, J. (2009). Wine Quality [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.24432/C56S3T>.

Variable	Mean	Median	SD	Min	Max	Shapiro-Wilks P-Value
citric.acid	0.319	0.310	0.145	0.000	1.660	0.000
residual.sugar	5.443	3.000	4.758	0.600	65.800	0.000
chlorides	0.056	0.047	0.035	0.009	0.611	0.000
free.sulfur.dioxide	30.525	29.000	17.749	1.000	289.000	0.000
total.sulfur.dioxide	115.745	118.000	56.522	6.000	440.000	0.000
density	0.995	0.995	0.003	0.987	1.039	0.000
pH	3.219	3.210	0.161	2.720	4.010	0.000
sulphates	0.531	0.510	0.149	0.220	2.000	0.000
alcohol	10.492	10.300	1.193	8.000	14.900	0.000

Identifying Key Characteristics of High-Quality Wine Using Clustering

Wines of very high (score of 9) and low quality (score of 3) have very few observations. To better identify characteristics of high-quality wines versus wines of lower quality, K Means Clustering was used to group the sample wines into 5 clusters of roughly equal size. Because quality follows a normal distribution, the clusters with the highest average quality score should have the highest portion of high-quality wine.

Clustering revealed that higher-quality vintages tend, on average, to have, higher levels of alcohol and lower levels of sugar, chlorides, and sulfates. To determine if the difference between the means of the lowest-quality and highest-quality clusters was statistically significant, a simple test was run across all of the variables. For all 11 independent variables, the cluster mean for both groups was significant at a alpha level of .01. Below is a summary table of the results by independent variables. The last column highlights some of the characteristics of high-quality Portuguese wine. Although we cannot extrapolate these results to all wine from all regions for the world, the last column can give consumers a good idea of what to look for when they are choosing between two or more wines.

Variable	T Statistic	P Value	High-Quality Wine Has...
fixed.acidity	3.713	2.195e-04	Lower Level Fixed Acids
volatile.acidity	2.958	3.198e-03	Lower Level Volatile Acids
citric.acid	4.985	8.337e-07	Lower Level Citric Acids
residual.sugar	22.917	2.079e-83	Lower Level of Sugar
chlorides	4.591	5.543e-06	Lower Level of Chlorides
free.sulfur.dioxide	24.885	7.220e-86	Lower Level of Free Surfer Dioxide
total.sulfur.dioxide	93.284	1.157e-287	Lower Level of Surfer Dioxide
density	30.984	2.302e-137	Lower density
pH	-2.857	4.414e-03	Higher pH
sulfates	4.096	4.619e-05	Lower Level of Sulfates
alcohol	-29.379	8.065e-141	Higher Level of Alcohol

Developing a Model to Predict the Quality of a Wine

Machine Learning Methods Used to Predict Quality

Given that there are more than two scores for quality, our dependent variable, this paper explored the use of K-Nearest Neighbors, Classification Trees, Random Forest, Support Vector Machines, and Artificial Neural

Network to predict the quality of the wine.

Cross Validation Methods Used to Evaluate the Each Method

To evaluate each method, the Validation Set method was used to measure the testing classification and error rates. In addition, Confusion Tables were used to analyze how effective each machine learning method was at scoring wine on the scale of 3 to 9.

Since the quality variable was ordinal, Mean Absolute Deviation (MAD) was also used to evaluate the accuracy of the machine learning methods. MAD preserves the order of the quality variable and measures how far off the predictions were when they missed the target. The lower the MAD value of a model, the more accurate the model was. MAD was calculated using the following formula.

$$MAD = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Cross Validation Results

Below is a table detailing the Cross Validation Results of all 5 prediction models. Overall, the models were fairly poor predictors of the actual quality score. On average, the models predicted the exact quality score 50% of the time. But even when they predicted the wrong score, they were generally in the ballpark. According to the Mean Absolute Deviations, most of the models were able to predict whether a wine was “good” or “bad,” even when they were unable to predict the exactly correct quality score. With the exception of the Artificial Neural Network model, all of the models were able to predict, on average, within 1 score of a wine’s actual ranking.

Model	Classification Rate	Error Rate	Mean Absolute Deviation
Random Forest	0.649	0.351	0.392
Artificial Neural Network	0.539	0.459	1.904
K-Nearest Neighbors	0.517	0.483	0.611
Classification Tree	0.514	0.486	0.547
Support Vector Machines	0.368	0.632	0.804

The Random Forest model with 2,423 trees and 3 variables considered at the end of each branch produced the lowest error rate of 35.1%. In the instances when the model incorrectly predicted a wine’s rank, the error only deviated on average by .392. In terms of variable importance, the Random Forest model’s top-three variables considered were alcohol, volatility acid, and citric acid, as measured by the Mean Decrease Gini.

The Random Forest model also did the best job of predicting across all wine quality scores. The Classification Tree, Support Sector Machine, and Artificial Neural Networks models failed to predict certain quality ranks. For instance, these models fail to rank certain wines into the lower or higher rankings — such as 3 to 4 or 7 to 9. Instead, they only gave wines the most common quality scores: 5 or 6.

Conclusion

The paper demonstrated that machine learning techniques could provide valuable insights into predicting wine quality based on chemical properties. Among the models tested, the Random Forest algorithm emerged as the most effective, achieving the highest classification accuracy and the lowest test error rate. Although the models were only modestly able to accurately predict the exact wine-quality scores, their ability to approximate a wine’s general quality (e.g., “good” or “bad”) was notable. This suggests that machine learning can serve as a useful tool to guide consumers in selecting high-quality wines.